

AIM TO TOP REVIEW AND TRAINING CENTER - ILIGAN
AREA 1 - COMPREHENSIVE EXAM

1. Also called the "Green Revolution" due to a large increase in crop yield and productivity.
- Agriculture 1.0
 - Agriculture 2.0
 - Agriculture 3.0
 - Agriculture 4.0

at 18 degrees working angle containing 25 disks spaced at 35cm? Disk diameter is 35 cm.

- 5.82 m
- 4.18 m
- 3.05 m
- None of the above

$N = 25$
 $S = 0.35$
 $D = .75$

2. Unbroken side of the furrow
- Backfurrow
 - Dead furrow
 - Furrow slice
 - Furrow wall

$$W = 0.95NS + 0.3D$$

$$W = 0.95(25)(35) + 0.3(35)$$

$$W = \underline{\hspace{2cm}}$$

(8.41)

3. It counteracts the pressure exerted by the furrow slice on the plow bottom.
- Share
 - Moldboard
 - Landside
 - Moldboard plow

8. Tillage operation which move soil to establish a desired soil elevation and slope
- Land forming
 - Land grading
 - land planning
 - earthmoving

4. Effective width of a cut of a single-bottom moldboard plow drawn by a power tiller is 15 cm. If machine is used to plow 3.5 ha farm, what would be the distance traveled?
- 133
 - 233
 - 333
 - 433

9. The fertility of a field is such that maximum corn yields are obtained with a population of 72,000 plants per hectare. The rows are 0.85 m apart and an average emergence of 65 percent is expected. How many seeds per hill should be planted if the hills are 0.7 m apart?

- 8
- 9
- 10
- 11

72,000 plants/ha
 $E = 65\%$
 $RS = 0.85m$
 $HS = 0.7m$
 6.59
 9.15

Area = Distance x Effective width

$$3.5 \text{ ha} \times \frac{10,000 \text{ m}^2}{1 \text{ ha}} = D \times 0.15 \text{ m}$$

D =

Expected (EPP)	Plant	Population
$= \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right) (\text{Seeds per hill}) (\text{Emergence})}{\text{Row Spacing} \times \text{Hill Spacing}}$		

5. The land is plowed in both directions side by side.
- Return plowing
 - Round-about plowing
 - Consecutive plowing
 - One-way plowing

100

$$\frac{000 \text{ plants}}{\text{ha}} = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right) (\text{Seeds per hill}) (0.65)}{0.85 \text{ m} \times 0.7 \text{ m}}$$

Seeds per hill =

6. Specially designed power tillers for wet land operation fitted with front-mounted cutting blades on a cage wheel.
- Rotavators
 - Floating power tillers
 - Moldboard plow
 - Tillers

10. In a sprayer, which component serves as a rigid conductor that delivers the spray from the hose to the nozzle?
- Lance
 - Boom
 - Wand
 - Pipe handle

7. What is the cutting width of single-disk harrow set

11. Walking in the rice field operating the sprayer with a spray swath of 1.5 m, a person covers 20 m/min. If

0.07 L/m²
0.17 m³/ha

nozzle discharge is 0.5 L/min, what is the application rate?

- 0.07 L/m² ($\frac{10,000 \text{ m}^2}{1 \text{ ha}}$)
Swath = 1.5 m
20 m/min
Q = 0.5 L/min
0.5 L/min = x (1.5 m) (20 m/min)
- a. 0.17 L/m²
 - b. 167 L/ha
 - c. 17 m³/ha
 - d. 2 L/ha

For sprayer: Q = Application rate
x Swath x Speed
0.5 L/min = Application rate x
1.5 m x 20 m/min
Application rate = _____

12. Which of the following statements is true regarding positive displacement pumps?

- a. They have an inverse relationship between discharge rate and pressure head.
- b. It discharges the same volume of water regardless of the head against which they operate.
- c. These are primarily used for high-flow applications.
- d. These are less efficient than centrifugal pumps at all operating conditions.

13. What is the theoretical maximum height that water can be lifted using suction?

- a. 2.31 ft
- b. 33 ft
- c. 3.3 m
- d. None of the above

14. In this system, harvesting and threshing are done mechanically.

- a. Traditional
- b. Conventional
- c. Mechanical
- d. Combined

15. What is the maximum separation loss for rice combine harvester?

- a. 0.5%
- b. 1.2%
- c. 1.3%
- d. 1.5%

16. What mechanism is employed in combines?

- a. Slicing/tearing action
- b. impact action
- c. scissor type action
- d. Two of the choices above

17. To gradually stop a fast-running tractor, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator

18. It transmits rotatory motion between parallel shafts.

- a. Bevel gears
- b. Spur gears
- c. Rivet gears
- d. Screw and drive

19. A tractor operator creates a maintenance checklist to ensure that oil changes, tire pressure checks, and air filter replacements are performed at regular intervals by all staff. Which 5S step does this represent?

- a. Seiton
- b. Seiketsu
- c. Seiso
- d. Shitsuke

20. The cylinder of a corn sheller will be driven by a 10-hp diesel engine using flat belt drive. The engine has a speed of 1720 rpm while corn sheller requires an engine speed of 720 rpm. The effective diameter of driver pulley is 12 inches. What is the require length of cross belt if center-to-center distance between pulleys is 60 in.

- a. 188 in
- b. 187 in
- c. 186 in
- d. 185 in

1720 rpm (12 in)
720 rpm

$$N_1 D_1 = N_2 D_2$$

$$720(D_1) = 1720(12)$$

$$D_1 = 28.67 \text{ inches}$$

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 + D_2)^2}{4C}$$

$$L = 2(60) + 1.57(28.67 + 12)$$

$$+ \frac{(28.67 - 12)^2}{4(60)}$$

$$L = \underline{\hspace{2cm}}$$

21. A 65m self-propelled combine makes an average stop of 5 minutes every time its 2-ton grain tank is to be unloaded. The yield of the 20-ha field is 30 tons. The operating speed is 4.0 kph. The time for turning

on the headland at the ends of the 400-m field is 15 seconds. What is the field efficiency?

- a. 83%
b. 84% 85.5
c. 85%
d. 86%

For TFC:

$$TFC = \frac{SWEff}{10}$$

$$TFC = \frac{(4 \text{ kph}) (5 \text{ m}) (1)}{10}$$

$$TFC = 2 \frac{\text{ha}}{\text{hr}}$$

For AFC:

$$\text{No. of rows} = 500 \text{ m} / 5 \text{ m/row} = 100 \text{ rows}$$

For t_1 : Actual working time in the rows at 100% efficiency

$$t_1 = 100 \text{ rows} \times \frac{400 \text{ m}}{\text{row}} \times \frac{\text{hr}}{4 \text{ km}} \times \frac{1 \text{ km}}{1000 \text{ m}} = 10 \text{ hr}$$

For t_2 : time for turning at headland

$$t_2 = 100 \text{ rows} \times \frac{1 \text{ turn}}{\text{row}} \times \frac{15 \text{ sec}}{\text{turn}} \times \frac{1 \text{ hr}}{3600 \text{ sec}} = 0.4167 \text{ hr}$$

For t_3 : time for unloading of tank

$$t_3 = 30 \text{ tons} \times \frac{1}{2 \text{ tons/tank}} \times \frac{5 \text{ min}}{\text{tank}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 1.25 \text{ hr}$$

$$T = t_1 + t_2 + t_3$$

$$T = 10 \text{ hr} + 0.4167 \text{ hr} + 1.25 \text{ hr} = 11.6667 \text{ hr}$$

$$AFC = \frac{A}{T} = \frac{20 \text{ ha}}{11.6667 \text{ hr}} = 1.71 \text{ ha/hr}$$

$$Eff = \frac{AFC}{TFC} = \frac{1.71 \text{ ha/hr}}{2 \text{ ha/hr}}$$

$$Eff = \underline{\hspace{2cm}}$$

22. What is the number of explosion of piston on a single cylinder, four stroke cycle engine, in 300 revolution of the camshaft?

- a. 100
b. 150
c. 300
d. 600

of explosion =

700 rev/min

2 cam = 1 crank

- a. Gasoline engine
b. Diesel engine
c. Compression ignition engine
d. AOTA

$$CR = \frac{(\frac{\pi(12 \text{ cm})^2}{4}(12 \text{ m})) + (300 \text{ cm}^3)}{300 \text{ cm}^3}$$

$$CR = (\frac{5.52}{1})$$

Diesel = 14-22
Gasoline = 4-12

23. This uses the engine's exhaust gases to spin a turbine, compressing the air entering the engine

- a. Naturally aspirated
b. Super charged
c. Turbo charged
d. Crankcase compressed

24. The engine has a clearance volume of 300 cm³, and both its bore and stroke measure 12 cm each. Based on these characteristics, this engine is most likely classified as a:

$$CR = \frac{CV + PD}{CV}$$

$$CR = 7.100 \text{ cm}^3$$

25. Changing of transmission oil should be done _____.

- a. every 3 months or 12000 km
b. every 6 months or 24000 km
c. every 9 months or 36000 km
d. every 12 months or 48000 km

26. Implement organic agriculture programs and projects approved by the NOAB.

- a. BAFPS
b. DOST-ATI
c. LGUs
d. Cooperative

27. According to RA 11203, how much is the annual appropriation for the Rice Fund?

- a. Php 10,000
- b. Php 100, 000
- c. Php 1, 000, 000
- d. Php 10, 000. 000, 000

28. The following are the reasons why mechanization in the Philippines is low EXCEPT _____.

- a. Issue about materials
- b. Issue about fabrication
- c. Lack of government support
- d. None of the above

29. If the revolution of the engine is not smooth, the possible reason is _____.

- a. Water in the fuel
- b. Spark plug
- c. Disconnected wires
- d. Excessive tire pressure

30. A person is working for 352 minutes in a farm harvesting. Estimate the power that can be generated by the person doing the job.

- a. 0.10 HP
- b. 0.11 HP
- c. 0.12 HP
- d. 0.13 HP

$$P = 0.35 - 0.092[\log (352 \text{ min})]$$

$$= \underline{\hspace{2cm}}$$

31. Which engine is not from Japan?

- a. Honda
- b. ACME
- c. Kubota
- d. Robin

32. What is thermal efficiency of diesel engine which uses 1.13 kg of fuel per kW-hr?

- a. 5%
- b. 6%
- c. 7%
- d. 8%

$$\text{Thermal efficiency} = \frac{\text{Useful energy output}}{\text{Energy provided by fuel}}$$

$$\text{Energy from fuel} = 1.13 \text{ kg} \times 45.46 \text{ MJ/kg} = 51.3698 \text{ MJ}$$

Useful energy output is stated "per kW-hr". Thus,
Energy output = 1 kW-hour = 3.6 MJ

$$\text{Thermal efficiency} = \frac{3.6 \text{ MJ}}{51.3698 \text{ MJ}} = 0.243 = \underline{\hspace{2cm}}$$

33. The ratio of reflected light to incoming light on a surface.

- a. Betz
- b. Albedo
- c. Lambert
- d. Lussac

34. In a sunny 12:00 noon in the Philippines, what is the likely voltage reading of solar pump without battery when load is in turned on?

- a. 20-22 V
- b. 16-19 V
- c. 14-15 V
- d. 11-13 V

35. With a wind velocity of 7.32 m/s, how much power is available if it is used for maximum electrical energy? Diameter of wind turbine is 4 m.

- a. 3081 W
- b. 1502 W
- c. 616 W
- d. 322 W

$$\text{Theoretical Power} = \frac{1}{2} \rho A V^3$$

$$\text{Theoretical Power} = \frac{1}{2} (1.25 \text{ kg/m}^3) \left(\frac{\pi (4 \text{ m})^2}{4} \right) (7.32 \text{ m/s})^3$$

$$\text{Theoretical Power} = \underline{\hspace{2cm}} \text{ Watts}$$

$$\text{Max Available Power} = \text{Theoretical P} \times \text{Power Coeff}$$

$$\text{Max Available Power} = 3080.51 \text{ W} \times 0.20$$

$$\text{Max Available Power} = \underline{\hspace{2cm}} \text{ W}$$

36. Pelton wheel is an example of _____.

- a. Impulse turbine
- b. Reaction turbine
- c. Cross-flow turbine
- d. Francis turbine

37. Which of the following got the highest manure available per day?

- a. Pigs
- b. Cow
- c. Buffalo
- d. horses

38. What does a low Viscosity Index (VI) indicate about an oil's viscosity with temperature changes?

- a. Greater change in viscosity with temperature
- b. Less change in viscosity with temperature
- c. No effect of temperature on viscosity
- d. Higher viscosity at all temperatures

low VI

little impact

- a. Cartesian coordinate system
- b. Universal Transverse Mercator (UTM)
- c. Geographic coordinate system (GCS)
- d. State Plane coordinate system

39. How does the spatial resolution of a remote sensing image affect the size of the grid cells?

- a. Higher spatial resolution results in larger grid cells
- b. Higher spatial resolution results in smaller grid cells
- c. Spatial resolution does not affect grid cell size
- d. Higher spatial resolution results in irregularly shaped grid cells

40. It is an active sensor that emits microwave signals and measures the energy reflected back.

- a. SAR
- b. LiDAR
- c. TIR
- d. Optical Imagery

41. A group of connected entities and activities that interact for a common purpose.

- a. System
- b. Information system
- c. Geographic
- d. Spatial

42. Which does not belong to the group?

- a. Hardware - *Physical*
- b. Software
- c. User - *Physical*
- d. Remote sensing - *Physical*

43. Which coordinate system is based on latitude and longitude?

44. Which GIS operation involves overlaying multiple spatial datasets to find relationships between them?

- a. Buffering
- b. Overlay analysis
- c. Spatial interpolation
- d. Network analysis

45. What tool would you use to calculate distances from a feature to all other features in the same layer?

- a. Buffer
- b. Near
- c. Select by Location
- d. Clip

46. At what intervals should load, speed of engine, fuel consumption, and other parameters be measured during the test of walking type agricultural tractor?

- A) Every 10 minutes
- B) Every 30 minutes
- C) Every 5 minutes
- D) Every 15 minutes

47. What is the minimum peak transmission efficiency required for the walking type tractor?

- A) 70% B) 75%
- C) 80% D) 85%

48. What should be the minimum plowing area for two-wheel tractor-drawn moldboard/ disc plows?

- A) 250 m² B) 500 m²
- C) 1,000 m² D) 2,000 m²

49. What is the minimum number of test trials that shall be conducted for the agricultural trailer?

- a. 1
- b. 2
- c. 3
- d. 4

50. What is the weeding efficiency of a weeder?

- A) 60%
- B) 70%
- ☒ C) 80%
- D) 90%

51. What will be the rated power of the electric motor replacement if the engine driver of a pump is 3 hp?

- A. 1 hp
 - ☒ B. 2 hp
 - C. 3 hp
 - D. 4 hp
- $3 \times \frac{2}{3} = 2$

52. The required milling degree for Rice Mills across all huller types both single and multi-pass.

- A. Well milled
- B. Brown rice
- ☒ C. Milled rice
- D. Paddy rice

53. The year AMTEC was established

- A. 1977
- B. 1982
- ☒ C. 1993
- D. 2013

54. The main leading agency in drafting standards.

- ☒ A. BAFS
- B. AMTEC
- C. PHILMech
- D. BAFE

55. The following are criteria for accreditation for accredited machinery testing centers, except;

- A. Must be an HEI, private or government entity
- B. Must have at least 2 Test Engineers
- C. Must have a quality assurance system
- ☒ D. Must have the building facility equipment and manpower

56. A farmer wishes to spray his cotton plantation at a rate of 300 gallons per hectare. Sixteen nozzles, 18 inches apart will be used. The forward speed of the tractor will be 5 kph. Compute the field capacity of the sprayer in hectare per hour.

- A. 5.6 ha/hr
 - ☒ B. 3.6 ha/hr
 - C. 4.7 ha/hr
 - D. 1.8 ha/hr
- $300 \text{ gal/ha} \times 16 \text{ nozzles} = 4800 \text{ gal/hr}$
 $4800 \text{ gal/hr} \div 10 \text{ gal/ha} = 480 \text{ ha/hr}$
 $480 \text{ ha/hr} \div 133.33 \text{ (from 18 inches)} = 3.6 \text{ ha/hr}$

57. Slenderness ratio is the ratio of

A. maximum size of column to minimum size of column

B. width of column to depth of column

☒ C. effective length of column to least radius of gyration of the column

D. effective length of column to width of column

58. A type of throw-in thresher which moves the materials being threshed in a helical manner around the threshing cylinder.

- A. axial-flow-thresher
- ☒ B. through-flow thresher
- C. hold-on thresher
- D. double drum thresher

59. What is defined as any tangible economic product that contributes directly or indirectly to the satisfaction of human want?

- A. Services
- ☒ B. Goods
- C. Commodities
- D. Goods or commodities

60. The office that was created thru Special Order No. 43 series of 2005 which serves as the technical staff of the Secretary in the administration and enforcement of the provisions of P.D. 1096 and its IRR.

☒ A. Department of Public Works and Highways

B. Bureau of Research and Standards

C. Bureau of Design

D. National Building Code Development Office

E. Bureau of Construction

61. A 6-m self-propelled combine makes an average stop of 3 min every time its 3-ton grain tank is to be unloaded. Gross yield of the 25-ha field is 50 tons. The operating speed is 5.5 kph. The time for turning on the headland at the ends of the 600-m field is

20 seconds. Find the actual field capacity:

- A. 3.0 ha/hr B. 4.0 ha/hr
C. 5.0 ha/hr D. 6.0 ha/hr

62. Given a two-row corn planter with a ground wheel diameter of 55 cm and row spacing of 70 cm, and assuming a seed plate with 30 cells, if the speed ratio of the ground wheel over the seed plate is 4.5, what is the plant population per hectare considering an average emergence rate of 90% and an average of 2 seeds per hill?

- A. 40,000 plts/ha
B. 50,000 plts/ha
C. 60,000 plts/ha
D. 70,000 plts/ha

63. If there are 4 million farms and a total agricultural land area of 20 million hectares, with an average of 6 farmers per farm, what is the collective human power contribution (in horsepower) in these farms?

- A. 0.3 hp
B. 0.6 hp
C. 0.9 hp
D. 1.2 hp

64. A 3.5-ha field was prepared using a 1.5-m floating power tiller. It took 25 hours to complete the operation at a speed of 3 kph. Calculate the field efficiency.

- A. 65% B. 70%
C. 75% D. 80%

65. Determine the tractor power required to operate a soil-working implement with four 16-inch moldboard plow bottoms at a depth of 12 inches and a working speed of 7 kph. Given a field efficiency of 85% and a specific draft of 10 lbs/inch², what is the necessary tractor power?

- A. 55 hp B. 65 hp
C. 75 hp D. 85 hp

66. Determine the number of tractors needed to plow a 10-ha field in one day (8 hrs/day) using a two-bottom moldboard plow at a speed of 6 kph. Each bottom has a width of 35 cm, and the field efficiency is 75%.

- A. 1 tractor B. 2 tractors
C. 3 tractors D. 4 tractors

67. Calculate the time (hours) required to plow a 6-ha field using a 14-cm animal-drawn plow at a speed of 3.5 kph and an efficiency of 80 percent.

- A. 85 hours B. 95 hours
C. 105 hours D. 115 hours

68. A 5-ha field was plowed using a 4X14-inch tractor-mounted moldboard plow. It took 12 hours to complete the operation at a tractor speed of 4.5 kph. What is the field efficiency?

- A. 70% B. 75%
C. 80% D. 85%

69. Determine the size of the diesel engine required for a 3 X 3 centrifugal pump. The pump has a discharge rate of 180 gpm, with a total head of 30 ft. The pump efficiency is 60%, and the V-belt efficiency is 90%. The air-cooled diesel engine delivery rating is 65%.

- A. 1.8 hp B. 2.5 hp
C. 3.3 hp D. 4.0 hp

70. A 5-m combine traveling at 4 kph can empty its 2-ton grain tank in 3 minutes. When unloading on-the-go, it achieves a field efficiency of 82%. If the field yield is 4 tons/ha, what would be the field efficiency if the combine stopped to load?

- A. 68% B. 72%
C. 76% D. 80%

71. Determine the drawbar horsepower required to pull a plow with 4X16-inch bottoms at the rate of 4 mph if the specific draft is 7 psi and depth of cut is 8 inches.

- A. 10.8 hp B. 15.5 hp
C. 21.2 hp D. 28.6 hp

72. You're tasked with selecting a diesel engine for a centrifugal pump measuring 4 inches X 4 inches. The pump needs to discharge 250 gallons per minute (gpm) against a total head of 30 feet. The pump's efficiency is 60%. The V belt efficiency is 90%. The air-cooled diesel engine's delivery rating is 65%. What size diesel engine should you

choose?

- A. 1.8 hp B. 2.6 hp
C. 3.5 hp D. 4.9 hp

73. A 5-meter combine operates at a speed of 6 kilometers per hour (kph). It can empty its 1.8-ton grain tank in 2.5 minutes while unloading on-the-go, achieving an 88% field efficiency. Given a field yield of 4 tons per hectare (ha), what would be the field efficiency if the combine stopped to load?

A. 72% B. 77%

C. 81% D. 86%

74. You're preparing a field measuring 50 meters X 100 meters using a 1.5-meter upland power tiller. Assuming a 15% overlap between passes and a 12-second turn-around time at the end of each pass, what is the theoretical field capacity?

A. 0.22 ha/hr B. 0.28 ha/hr

C. 0.36 ha/hr D. 0.42 ha/hr

75. Calculate the theoretical field capacity of a 2.5-meter combine traveling a distance of 120 meters in 60 seconds.

A. 0.04 ha/hr B. 0.62 ha/hr

C. 1.35 ha/hr D. 1.92 ha/hr

76. This Act, consisting of 16 sections, establishes the State policy to promote the competitiveness of the sugarcane industry and maximize the utilization of sugarcane resources, and improve the incomes of farmers and farm workers, through improved productivity, product diversification, job generation, and increased efficiency of sugar mills.

A. RA 2207 B. RA 10659

C. RA 11203 D. RA 11598

E. RA 11901

77. This provides the necessary policy framework, institutional mechanisms and mandate to the local government units (LGUs) to achieve 25% waste reduction through establishing an integrated solid waste management plan based on 3Rs (reduce, reuse and recycling).

- A. DENR DAO 2019-21
B. NSWMC Resolution No. 1363 Series of 2020
C. RA 9003 - ecological solid waste
D. PD 825
E. None of the Above

78. Department of Agriculture's Department Circular No. 1 Series of 2013 contains the:

A. Implementing Rules and Regulations for RA NO. 10601, Otherwise known as the "Agricultural and Fisheries Mechanization Law".

B. Implementing Policies, Guidelines and Procedures for the Establishment and Operations of the Agricultural and Fisheries Mechanization Research, Development and Extension Network (AFMECHRDEN)

C. National Guidelines on Testing and Evaluation of Agricultural and Fisheries Machinery

D. Implementing Guidelines on the Strengthening and Establishment of the Agricultural and Biosystems Engineering Groups of the Local Government Units (LGUs)

E. None of the above

79. The Testing and Evaluation Accreditation Advisory Committee (TEAAC) was tasked to review and recommend the granting of Certificate of Accreditation to Testing Centers and Test Engineers, granting of Certificate of Conformity, and the revocation, suspension and reinstatement of such if the need arises. Which of the following

is not a member of the committee?

A. BAFS B. PCAF

C. PRB-ABE D. PSABE E. PhilRice

80. Document given to machinery owners after they complied with all the documentary requirements and have paid the registration fee for machinery registration.

A. Permit to Operate Registration (PTO)

- B. Certificate of Agricultural and Fisheries Machinery and Equipment Registration (CAFMER)
- C. Certificate of Equipment and Machinery Registration (CEMR)
- D. Certificate of Accreditation (CoA)
- E. Certificate of Conformity (CC)

81. This must be high enough in order to self-ignite the air-fuel mixture in a diesel engine.

- A. compression temperature
- B. intake pressure
- C. valve compression
- D. engine pressure
- E. fuel temperature

82. Which pair is mismatched?

- A. indirect cooling - water-cooled engine
- B. spark ignition - gasoline engine
- C. Otto cycle - gasoline engine
- D. constant volume process - diesel engine
- E. direct cooling - air-cooled engine

83. The indicated horsepower of a tractor engine is 300 hp. What is its expected power takeoff?

- A. 245 hp B. 257 hp
- C. 261 hp D. 225 hp E. 294 hp

84. Compute the power generated by a multi-blade windmill with a rotor diameter of 6.0 m. The average wind velocity is 10.95 m/s. Energy conversion efficiency is 29.95% and power coefficient is 0.15. Assume density of air is 1.2 kg/m³.

- A. 0.8 kW B. 1.0 kW
- C. 1.2 kW D. 1.4 kW

85. Convert the binary number to decimal: 1101110012

- A. 44110 B. 13410
- C. 41810 D. 54110

86. Which of the following is not an RCEF implementing agency?

- A. Philippine Center for Postharvest Development and Mechanization

- B. Bureau of Plant Industry
- C. Philippine Rice Research Institute
- D. Landbank of the Philippines
- E. International Rice Research Institute

87. A document that companies use to outline the products or services, pricing and time frame they can offer clients for completing a specific project

- A. Canvass Form
- B. Bill of Materials
- C. Bid Proposals
- D. Feasibility Study
- E. All of the above

88. It refers to ratio of the kinetic energy of air that a wind rotor can convert into mechanical energy over the theoretical maximum available from the wind.

- A. Power coefficient
- B. Torque coefficient
- C. Betz coefficient
- D. Speed coefficient
- E. none of the above

89. In an ideal Diesel engine, the combustion of fuel occurs at :

- A. constant pressure
- B. constant temperature
- C. constant volume
- D. constant enthalpy
- E. none of the above

90. These are the type of middleman who do the business of negotiating in behalf of the producers and the buyers. They do not assume ownership of the products that they handling.

- A. Retailers X
- B. Wholesalers X
- C. Assemblers
- D. Agents and brokers
- E. none of the above

91. It is a device used to reduce the toxicity of emissions from an internal combustion engine. It works by providing an environment for a chemical reaction wherein toxic combustion by-products are converted to less-toxic substances.

- A. Muffler
- B. Header
- C. Turbocharger

- D. catalytic converter
E. none of the above

92. A 85 mm × 85 mm engine runs at 2,000 rpm. What is the maximum piston speed inside the cylinder?

- A. 105 m/s B. 17.8 m/s
C. 25 m/s D. 75 m/s

93. A diesel engine was tested for a continuous 3 hours of operation. It consumed at total of 5 liters and developed a constant power of 7.5 kW. If the fuel contains 45.46 MJ/kg, what is the approximate thermal efficiency?

- A. 40% B. 35%
C. 33% D. 25%
E. none of the above

94. Which one of the following statements is correct?

A. The number of years required to recover the initial cash investment in a project, is called Pay Back period (PBP).

B. The discount rate that equates the present value of the expected Net Cash Flows (CFs) with the Initial-Cash

Outflow (ICO) is known as internal rate of return.

C. The present value of the proposal's net cash flows, less the proposal's initial cash outflow is known as the Net Present Value (NPV)

E. All of these

95. The ratio of current assests to current liabilities is known as

- A. Liquidity ratio
B. Current ratio
C. Acid-Test (or Quick) ratio
D. Debts ratio

96. The recommended application rate of a certain chemical is 5 kg/ha. The calibration test of a 20-liter knapsack sprayer is as follows: Nozzle discharge = 1.5 liters/min, swath = 1.8 m, and walking speed = 25 m/min. Determine the amount of chemicals to mix with water per loading:

- A. 110 gms/load
B. 160 gms/load
C. 220 gms/load

D. 280 gms/load

97. The application rate of a certain chemical is recommended at 6 kg/ha. The calibration test of a 25-liter knapsack sprayer is as follows: Nozzle discharge = 1.8 liters/min, swath = 2.0 m, and walking speed = 30 m/min.

Determine the total volume of water applied per hectare:

- A. 250 liters B. 350 liters
C. 450 liters D. 550 liters

98. A 5-m combine traveling at 6 kph can empty its 2-ton grain tank in 3 minutes. When unloading on-the-go, it has an

80% field efficiency. Field yield is 4.5 tons/ha. What would be the field efficiency if the combine stopped to

- load?
A. 70% B. 75%
C. 80% D. 85%

99. What is the theoretical field capacity of a 4-m combine traveling at 120 meters in 60 seconds?

- A. 1.0 ha/hr B. 1.5 ha/hr
C. 2.0 ha/hr D. 2.5 ha/hr

100. A 6-m self-propelled combine makes an average stop of 3 min every time its 3-ton grain tank is to be unloaded. Gross yield of the 25-ha field is 50 tons. The operating speed is 5.5 kph. The time for turning on the headland at the ends of the 600-m field is 20 seconds. Find the theoretical field capacity of the combine:

- A. 2.5 ha/hr B. 3.5 ha/hr
C. 4.5 ha/hr D. 5.5 ha/hr

Bayaw, remember that each hour you dedicate each page you study, and every challenge you face brings you closer to your dream.

One day, all this hard work will be worth it!